Gantavya Bhatt

Machine Learning, Optimization and Data Interpretation Lab
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Research Interests

My key interest lies in data-efficient multimodal model pipelines, in particular, how can we find the most influential datapoints for our tasks at hand? I extensively use the Submodular Optimization framework to find core-sets.

Education

University of Washington, Seattle *2021 - 2026 (Expected)*
Ph.D Candidate
GPA: 3.87/4.0
Advisor: Prof. Jeff A. Bilmes
Thesis: Towards the Empirical Aspects of Submodualrity

Indian Institute of Technology Delhi *2016 - 2020*
B.Tech in Electrical Engineering
GPA: 9.141/10
Advisor: Prof. Sumeet Agarwal

Work Experience

NVIDIA ADLR *June - Dec, 2025*
Research Scientist Intern
Advisors: Sanjeev Satheesh, Mostofa Patwary
Project: Curriculum learning for pertaining LLMs.

NVIDIA ADLR *June - Nov, 2024*
Research Scientist Intern
Advisors: Kevin Shih, Rafael Valle
Project: Improving the video-to-audio generation pipeline.

Amazon AWS Rekognition *June - September, 2023*
Applied Scientist Intern
Advisors: Achin Jain, Yang Zou, Tiffany Deng
Project: Alignment of text-to-image models with supervised fine-tuning

Publications

Preprints and Technical Reports

*, † - equal contribution

2. **Coresets Before Score Sets: Evaluation-Unsupervised Prompt Subset Selection for LLM Benchmarks**
Jihan Yao*, **Gantavya Bhatt***, Arnav Das*, Peter Jin, Ke Bao, Qiaolin Yu, Khushi Bhardwaj, Chang Su, Jialei Wang, Yikai Zhu, Sugam Devare, Damon Mosk-Aoyama, Zhen Dong, Venkat Krishna Srinivasan, Yineng Zhang, Oleksii Kuchaiev, Jiantao Jiao, Banghua Zhu, Jeff Bilmes
arXiv preprint, July 2026
***Summary**: We study evaluation-unsupervised benchmark coreset selection using submodular objectives.*

1. **How Much Is a Dataset Worth? Scaling Laws, the Vendi Score, and Matrix Spectral Functions**
Jeff A. Bilmes, **Gantavya Bhatt**, Arnav M. Das
Under review
***Summary**: We show a missing information axis in scaling laws and introduce a new class of submodular functions.*

Accepted Publications and Technical Reports

21. **Nemotron 3.5 Super and Lightning: Open, Efficient Mixture-of-Experts Hybrid Mamba-Transformer Models for Agentic AI**
NVIDIA; **Gantavya Bhatt** and collaborators (full list in report).
NVIDIA Technical Report, 2026
***Summary**: Open, efficient Nemotron 3.5 Super and Lightning models for agentic AI.*

20. **Benchmarking Single-Factor Physical Video-to-Audio Generation**
Tingle Li, Siddharth Gururani, Kevin J. Shih, **Gantavya Bhatt**, Sang-gil Lee, Zhifeng Kong, Arushi Goel, Gopala Anumanchipalli, Ming-Yu Liu
Poster- *Conference on Computer Vision and Pattern Recognition (**CVPR**) 2026*
***Summary**: We create a physics based benchmark for evaluating Video-to-Audio generative models.*

19. **Nemotron 3 Ultra: Open, Efficient Mixture-of-Experts Hybrid Mamba-Transformer Model for Agentic Reasoning**
NVIDIA; **Gantavya Bhatt** and collaborators (full list in report).
NVIDIA Technical Report, 2026
***Summary**: An open hybrid Mamba-Transformer mixture-of-experts model for agentic reasoning.*

18. **Nemotron 3 Super: Open, Efficient Mixture-of-Experts Hybrid Mamba-Transformer Model for Agentic Reasoning**
NVIDIA **Pretraining Data Contributor**: Abhinav Khattar, Aleksander Ficek, ... , **Gantavya Bhatt**, ... , Yejin Choi, Yev Meyer, Ying Lin, Zihan Liu.
NVIDIA Technical Report, 2026
***Summary**: We present Nemotron 3 Super 122B-A12B, a Mixture-of-Experts hybrid MambaTransformer language model.*

17. **Matched Data, Better Models: Target Aligned Data Filtering with Sparse Autoencoders**
Arnav M. Das*, **Gantavya Bhatt***, Sahil Verma, Yiping Wang, Viswa Virinchi Muppirala, Jeff Bilmes
*The Fourteenth International Conference on Learning Representations (**ICLR**) 2026*
***Summary**: We scale submodularity for target-aware data selection for pretraining data using Sparse Autoencoders.*

16. **NVIDIA Nemotron 3: Efficient and Open Intelligence**
NVIDIA **Pretraining Data Contributor**: Abhinav Khattar, Aleksander Ficek, ... , **Gantavya Bhatt**, ... , Yejin Choi, Yev Meyer, Ying Lin, Zihan Liu.
NVIDIA Technical Report, 2025
***Summary**: We present the NVIDIA Nemotron 3 family of open hybrid Mamba-Transformer mixture-of-experts models.*

15. **Nemotron 3 Nano: Open, Efficient Mixture-of-Experts Hybrid Mamba-Transformer Model for Agentic Reasoning**
NVIDIA **Pretraining Data Contributor**: Abhinav Khattar, Aleksander Ficek, ... , **Gantavya Bhatt**, ... , Yejin Choi, Yev Meyer, Ying Lin, Zihan Liu.
NVIDIA Technical Report, 2025
***Summary**: We present Nemotron 3 Nano 30B-A3B, a Mixture-of-Experts hybrid MambaTransformer language model.*

14. **How Many Images Does It Take? Estimating Imitation Thresholds in Text-to-Image Models**
Sahil Verma, Royi Rassin, Arnav Das*, **Gantavya Bhatt***, Preethi Seshadri*, Chirag Shah, Jeff Bilmes, Hannaneh Hajishirzi, Yanai Elazar
*Transactions on Machine Learning Research (**TMLR**) 2025*
***NeurIPS 2024** Workshop on Attributing Model Behavior at Scale, Regulatable ML, Safe Gen AI*
***EMNLP 2024** 6th Workshop on Natural Legal Language Processing (NLLP)*
***Summary**: We aim to find the Number of Images Required by a Text-to-Image Models to Imitate Something*

13. **Comparing Bad Apples to Good Oranges: Aligning Large Language Models via Joint Preference Optimization**
Hritik Bansal*, Ashima Suvarna*, **Gantavya Bhatt***, Nanyun Peng, Kai-Wei Chang, Aditya Grover
***Findings (Long) -** The 63rd Annual Meeting of the Association for Computational Linguistics, ACL'2025*
***ICML 2024** Workshop on DMLR, Models of Human Feedback for AI Alignment*
***Summary**: We propose optimizing joint preference over arbitrary instruction response pairs instead of conditional.*

12. **COBRA: COmBinatorial Retrieval Augmentation for Few-Shot Adaptation**
Arnav Das*, **Gantavya Bhatt***, Lilly Kumari, Sahil Verma and Jeff Bilmes
Poster- *Conference on Computer Vision and Pattern Recognition (**CVPR**) 2025*
***ICML 2024** Workshop on Data-Centric Machine Learning Research*
***Summary**: We generalize retrieval augmented learning using Submodular Mutual Information Measures.*

11. **Deep Submodular Peripteral Networks**
Gantavya Bhatt*, Arnav Das* and Jeff Bilmes
Spotlight- *Thirty-eighth Conference on Neural Information Processing Systems (**NeurIPS**) 2024*
***Summary**: We propose a framework to learn deep submodular functions effective at Streaming summarization.*

10. **Effective Backdoor Mitigation in Vision-Language Models Depends on the Pre-training Objective**
Sahil Verma, **Gantavya Bhatt**, Avi Schwarzschild, Soumye Singhal, Arnav Das, Chirag Shah, John P Dickerson, Pin-Yu Chen, Jeff Bilmes

*Transactions on Machine Learning Research (**TMLR**) 2024* 🏆 Journal to Conference Certification
Best Oral Award 🏆 - *Workshop on Backdoors in Deep Learning at **NeurIPS** 2023*
***Summary**: We showcase shortcoming of SOTA backdoor mitigation strategy CleanCLIP.*

9. **An Experimental Design Framework for Label-Efficient Supervised Finetuning of Large Language Models**
Gantavya Bhatt*, Yifang Chen*, Arnav Das*, Jifan Zhang*, Sang T. Truong, Stephen Mussmann, Yinglun Zhu, Jeff Bilmes, Simon Shaolei Du, Kevin Jamieson, Jordan P. Ash, Robert D Nowak
***Findings (Long)** - The 62nd Annual Meeting of the Association for Computational Linguistics, ACL'2024*
Best Poster Award 🏆 - *Midwest Machine Learning Symposium (MMLS) 2024*
***Summary**: We propose an experimental design based framework for Label-Efficient Instruction tuning of Large Language Models.*

8. **LabelBench: A Comprehensive Framework for Benchmarking Adaptive Label-Efficient Learning**
Jifan Zhang*, Yifang Chen*, Gregory Canal, Arnav M. Das†, **Gantavya Bhatt**†, Stephen Mussmann, Yinglun Zhu, Jeffrey Bilmes, Simon S. Du, Kevin Jamieson, Robert D. Nowak
Journal of Data-centric Machine Learning Research (DMLR), March 2024
Poster- *Workshop on Real world Machine Learning at **NeurIPS** 2023.*
***Summary**: We benchmark Active Learning algorithms in conjunction with SSL in the Vision-Language Models.*

7. **Accelerating Batch Active Learning Using Continual Learning Techniques**
Arnav Das*, **Gantavya Bhatt***, Megh Bhalerao, Vianne Gao, Rui Yang and Jeff A. Bilmes
*Transactions on Machine Learning Research (**TMLR**) 2023*
Poster- *Workshop on Data Centric Machine Learning at ICML'23*
***Summary**: We propose a Continual Learning approach to alleviate a major bottleneck in Active Learning.*

6. **RadarHD: Demonstrating Lidar-like Point Clouds from mmWave Radar**
Akarsh Prabhakara, Tao Jin, Arnav Das, **Gantavya Bhatt**, Lilly Kumari, Elahe Soltanaghaei, Jeff Bilmes, Swarun Kumar, Anthony Rowe
Poster- *ACM MobiCom 2023*
***Summary**: We propose an algorithm to predict Point Clouds given mmWave Radar data.*

5. **High Resolution Point Clouds from mmWave Radar**
Akarsh Prabhakara, Tao Jin, Arnav Das, **Gantavya Bhatt**, Lilly Kumari, Elahe Soltanaghaei, Jeff Bilmes, Swarun Kumar, Anthony Rowe
Poster- *International Conference of Robotics and Automation (**ICRA**) 2023*
***Summary**: We propose an algorithm to predict Point Clouds given mmWave Radar data.*

4. **Matryoshka Representation Learning**
Aditya Kusupati*, **Gantavya Bhatt***, Aniket Rege*, Matthew Wallingford, Aditya Sinha, Vivek Ramanujan, William Howard-Snyder, Kaifeng Chen, Sham Kakade, Prateek Jain and Ali Farhadi
Poster- *Thirty-sixth Conference on Neural Information Processing Systems (**NeurIPS**) 2022*
Oral- *Vision Transformers: Theory and Applications workshop @ **NeurIPS**, 2022.*
Oral- *Self-Supervised Learning - Theory and Practice workshop @ **NeurIPS**, 2022*
***Summary**: We propose an algorithm to learn nested representation for adaptive/elastic deployment and fast retrieval.*

3. **Systematic Generalization in Neural Networks-based Multivariate Time Series Forecasting Models**
Hritik Bansal*, **Gantavya Bhatt***, Pankaj Malhotra and Prathosh AP
Oral** - 2021 International Joint Conference on Neural Networks **(IJCNN) 2021
***Summary**: We show the utility of disentangled architecture for robust time-series forecasting in dynamical system.*

2. **Tighter m-DPP Coreset Sample Complexity Bounds**
Gantavya Bhatt and Jeffery A. Bilmes
***Spotlight** - 2021 International Conference of Machine Learning **(ICML)** workshop on SubSetML.*
***Summary**: We show that the chaining can be extended to the DPP sampling procedure to improve complexity bounds.*

1. **How much complexity does an RNN architecture need to learn syntax-sensitive dependencies?**
Gantavya Bhatt*, Hritik Bansal*, Rishubh Singh* and Sumeet Agarwal
Poster - *58th Annual Conference of Associations of Computational Linguistics (**ACL**), Student Research Workshop, 2020*
***Summary**: We propose a brain-inspired recurrent network called - "Decay RNN" as good as LSTM for linguistic tasks.*

Peer reviewed abstracts

1. **Can RNNs trained on harder subject-verb agreement instances still perform well on easier ones?**
Hritik Bansal*, **Gantavya Bhatt*** and Sumeet Agarwal.
Poster - *In Proceedings of the Society for Computation in Linguistics **(SCiL)** 2021.*

Professional Responsibilities

- **Program Committee:** AAAI'21, ACL-IJCNLP'21, ICML'21 (SubSetML), AAAI'22, ICLR'22, NeurIPS'22 (*****), ICML'23, NeurIPS'23(*****) , ICLR'24, ICML'24, NeurIPS'24, ICLR'25, ICML'25, NeurIPS'25, ICLR'26 * - top-reviewer

Invited Talk(s)

- Data Efficiency in Machine Learning. *NVIDIA ADLR*
- Deep Submodular Peripteral Networks. *WAIL Talks @ UWCSE*
- Deep Submodular Peripteral Networks. *ML-OPT Talks @ UWCSE*
- Accelerating Batch Active Learning Using Continual Learning Techniques. *RAIVN Lab @ UWCSE*
- Discerning OOD Generalization in RNN LMs using Agreement Attraction Effects. *New York University*

Selected Awards and Honors

- Top reviewer for **NeurIPS'23** 2023
- Top reviewer for **NeurIPS'22** 2022
- University of Washington graduate travel grant award for NeurIPS'22. 2022
- Teaching Assistant award for undergraduate course on Linear Algebra and Differential equations (MTL101). 2020
- Secured position in the **Merit Top 7%** of **IIT Delhi** 3 times for **1st, 2nd and 6th** semester 2020
- One in **top 20 students across the globe** selected for the **EECS Research Camp** by the **HKUST** 2019
- Achieved **All India Rank 830** among **1.2 million** candidates appearing in Joint Entrance Examination 2016
- Awarded KVPY Fellowship from Government of India and IISc Bengaluru - All India Rank 443. *2015*

Teaching

- EE-516: Speech Recognition Systems. University of Washington. Fall 2026.
- EE-511: Advanced Introduction to Machine Learning. University of Washington. Wi2025, Wi2026.
- ELL-888: Graduate Deep Learning. IIT Delhi. Spr 2020.
- MTL-101 Introduction to Linear Algebra. IIT Delhi. Spr 2020.

Mentoring

- Tianyi Zhang (UW CSE, MS) - Datamixture optimization w/ Submodular Functions (co-advised with Arnav Das). 2026-
- Paula Cortes (UW Linguistics, MS) - Information Retrieval under different perspectives. 2025-2026.

References

- Jeff Bilmes, Professor, University of Washington
- Sanjeev Satheesh, Research Scientist, NVIDIA
- Kevin Shih, Research Scientist, NVIDIA
- Rafael Valle, Research Scientist, Meta
- Mostofa Patwary, Director of LLM Training, NVIDIA